PHONE • 16 OCTOBER 2026
AI • BAYESIAN OPTIMIZATION • US EQUITIES

Dave’s Daily

tldr

1. Meta released Muse Spark Open on 15 October: 816 billion parameters, 42 billion active on each token. On Meta’s StrongREJECT card the score is 0.5, against 62.9 to 86.4 for the other open models listed.

2. Eight states sued OpenAI on 14 October. The complaints allege that ChatGPT harmed children and misled adults, and they ask courts to change the product.

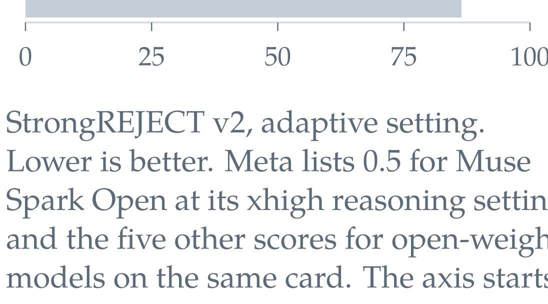
3. A Shanghai group turns a language model’s rankings into the geometry of a Bayesian optimization search. In a lab, 34 measurements of an adhesive cleared four engineering thresholds.

AI

On 15 October Meta released the weights for Muse Spark Open. The model card describes a mixture-of-experts transformer: 816 billion parameters in total, 256 experts, 8 experts chosen per token, so about 42 billion parameters do the work of each token. The context window is 1 million tokens. Inputs can be text, image, video, or audio. The output is text. The license allows commercial use. Axios reports that Meta and Apple ran the model locally, with Apple’s MLX, on two or four Mac Studios with M5 Ultra chips. The model card’s own deployment note is two of those machines. Alexandr Wang told Axios that, because the model runs on the machine, Meta does not see the prompts.

The chart is StrongREJECT v2 in the adaptive setting, a score Meta says should be lower. Muse Spark Open is at 0.5. GLM 5.3 is at 62.9, Qwen3.8 at 65.6, Kimi K3 at 66.6, Nemotron 3 Ultra at 80.7, and Inkling at 86.4. The same card says the released file refused 99.2 percent of jailbreaks on an internal cyber test, and that a modified copy can lose that behavior.

The same two days brought a separate fight over what a deployed chatbot does to people. On 14 October, attorneys general in Arizona, Arkansas, Idaho, Iowa, Kansas, Montana, Nebraska, and West Virginia sued OpenAI in their own state courts (AP; The Guardian). AP, updating the story on 15 October, says the suits claim ChatGPT violated consumer-protection laws by hurting children and deceiving adults. The states want changes to how the product works, and they want penalties. These are allegations in complaints, not findings.



StrongREJECT v2, adaptive setting. Lower is better. Meta lists 0.5 for Muse Spark Open at its xhigh reasoning setting, and the five other scores for open-weight models on the same card. The axis starts at zero and ends at 100. Source: Meta Superintelligence Labs, Muse Spark Open model card, 15 October 2026.

Markets

arXiv:2610.20104, posted 14 October, is a controlled study of transformers trained on limit-order-book messages. Fernando Acero, at J.P. Morgan QTR AI Research, trains GPT-style models on simulators whose hidden settings are known, then asks how close the model is to the best possible next-message forecast.

On the simplest simulator, where that forecast can be computed exactly, the trained model matches it to 3.4×10^{-4} nats per token at 512 messages of context. The bootstrap 95 percent interval on the gap is $[-7.3, +0.2] \times 10^{-4}$. On three harder simulators the model reaches the oracle floor, including one whose hidden state drifts inside an episode, so matching the oracle requires tracking a moving regime.

Scaling fits hit a floor within 2 millinats of the known entropy, and models of about 10^5 parameters already reach it. Holding the token count fixed and changing the number of distinct market regimes does change the result: a model trained on one regime scores 3.57 nats against a 1.34 floor on fresh regimes, and the gap closes only near 256 regimes. On one day of NASDAQ order flow, from the LOBSTER reconstruction of the ITCH feed, zero-shot transfer depends on whether the simulator mix covers real flow. Adding the heavy-tailed family nearly closes that gap, and the pretrained model beats training from scratch at every budget on three held-out tickers.

Bayesian optimization

arXiv:2610.14626, posted 10 October, is *TagBO: LLM-Driven Task-Aware Graph Bayesian Optimization for Scientific Discovery*, by Xinzhe Yuan and Zimu Zeng, with Zhuo Chen, Jianshu Zhang, Jinzong Dong, Jingyi Chen, Nanyang Ye, Huan Xiong, and Qinying Gu, at the Shanghai Artificial Intelligence Laboratory, Harbin Institute of Technology, Fudan University, Shanghai Jiao Tong University, and collaborating labs.

A language model asked for a score on every candidate is noisy. TagBO asks for ranks. For each categorical choice, the model names a few factors that should matter for the task and ranks the candidates on each factor. Similar rank patterns become edges in a graph. A second graph connects candidates that look alike in ordinary chemical descriptors. The two graphs are blended into a representation, and a constrained Gaussian process searches that representation together with the continuous knobs. The language model shapes which candidates count as neighbors. It does not pick the next experiment by itself.

The lab test is a thermally conductive adhesive. The search picks 10 fillers from 29 materials and sets 10 continuous amounts, 20 variables in all. Each physical measurement took about three days. After 34 measurements, thermal conductivity had moved from 0.1823 to 2.995 watts per meter-kelvin, above a qualification line of 2.8. The same formula measured 3.36 MPa in tension (the line was 1), 2.92 MPa in shear (the line was 2), and 23.5 kilovolts per millimeter in dielectric strength (the line was 7.9). A usual TuRBO start on a 20-variable problem spends 40 evaluations before the search proper. TagBO’s whole budget here was smaller than that start. The paper does not run TuRBO on this adhesive, so the comparison is to the size of the usual opening design.

On six chemistry and materials benchmarks, including a Suzuki–Miyaura yield search over 384 ligand, base, and solvent combinations, the authors report fewer evaluations to reach 95 percent of the best known value than one-hot Bayesian optimization, descriptor-only search, COMBO, BOPRO, and LGBO. Larger language models reach that line sooner. The early search moves similarly across model sizes, which the authors read as the coarse ranks doing the work.

Updated 16 October 2026. Phone and letter match. Sources: Meta and Axios, 15 October; AP, 15 October; arXiv, 14 October and 10 October. The state lawsuits are allegations. Created with malvin.